API Documentation

1. Overview

- **Project**: Cities API Coursework Project
- **Production Base URL**: `https://webservice-cw1.onrender.com`
- **Local Base URL**: `http://127.0.0.1:8000`
- **Format**: JSON (except `204 No Content`)
- **Interactive Docs**:
  - Swagger UI: `/docs`
  - ReDoc: `/redoc`

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2. Authentication

This API currently does **not** require authentication.

- No token, API key, or session is required.
- If authentication is added later, this section should be updated with:
  - Auth type
  - Header format
  - Token lifecycle

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3. Error Codes

Common status codes used by this API:

- `200 OK`: successful request
- `201 Created`: resource created
- `204 No Content`: successful deletion
- `404 Not Found`: target resource does not exist
- `409 Conflict`: duplicate resource or ambiguous query
- `422 Unprocessable Entity`: request validation failed (FastAPI/Pydantic)
- `500 Internal Server Error`: unexpected server failure

Example error response:

```
{
  "detail": "city not found"
}
```

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4. Endpoints

4.1 Health

`GET /health`

- **Purpose**: service health check
- **Parameters**: none
- **Response (200)**:

```
{
  "status": "ok"
}
```

Example:

```
curl "https://webservice-cw1.onrender.com/health"
```

4.2 Cities

`POST /cities`

- **Purpose**: create a city record
- **Body**:
  - `station_id` (string)
  - `city_name` (string)
  - `country` (string)
  - `state` (string)
  - `iso2` (string, length 2)
  - `iso3` (string, length 3)
  - `latitude` (number)
  - `longitude` (number)
- **Responses**:
  - `201` : created object

- `409`: `station_id` already exists
- `422`: invalid body format

Example request:

```
curl -X POST "https://webservice-cw1.onrender.com/cities" \
  -H "Content-Type: application/json" \
  -d '{
    "station_id":"T001",
    "city_name":"Test City",
    "country":"Testland",
    "state":"Test State",
    "iso2":"TT",
    "iso3":"TST",
    "latitude":10.1234567,
    "longitude":20.1234567
  }'
```

Example response (201):

```
{
  "station_id": "T001",
  "city_name": "Test City",
  "country": "Testland",
  "state": "Test State",
  "iso2": "TT",
  "iso3": "TST",
  "latitude": 10.1234567,
  "longitude": 20.1234567
}
```

`GET /cities`

- **Purpose**: list cities with pagination
- **Query params**:
  - `skip` (int, default `0`, min `0`)
  - `limit` (int, default `20`, min `1`, max `200`)
- **Response**: `200` + city array

Example:

```
curl "https://webservice-cw1.onrender.com/cities?skip=0&limit=5"
```

`GET /cities/by-latitude`

- **Purpose**: list cities sorted by latitude descending
- **Query params**:
  - `limit` (int, default `20`, min `1`, max `300`)
- **Response**: `200` + city array

Example:

```
curl "https://webservice-cw1.onrender.com/cities/by-latitude?limit=10"
```

`GET /cities/nearest`

- **Purpose**: find nearest city by input city name
- **Query params**:
  - `city_name` (string, required)
  - `iso2` (string, optional, disambiguation)
- **Responses**:
  - `200`: nearest city payload
  - `404`: source city not found
  - `409`: multiple cities found without disambiguation

Example:

```
curl "https://webservice-cw1.onrender.com/cities/nearest?city_name=London&iso2=GB"
```

Example response (200):

```
{
  "query": {
    "city_name": "London",
    "iso2": "GB"
  },
  "source_city": {
    "station_id": "03772",
    "city_name": "London",
    "country": "United Kingdom",
    "state": "England",
    "iso2": "GB",
    "iso3": "GBR",
    "latitude": 51.5072222,
    "longitude": -0.1275
  },
  "nearest_city": {
    "station_id": "XXXXX",
    "city_name": "Nearby City",
    "country": "United Kingdom",
    "state": "England",
    "iso2": "GB",
    "iso3": "GBR",
    "latitude": 51.4,
    "longitude": -0.2
  },
  "distance_km": 12.345
}
```

`GET /cities/{station_id}`

- **Purpose**: get one city by station ID
- **Path param**:
  - `station_id` (string)
- **Responses**:
  - `200` : city object
  - `404` : city not found

Example:

```
curl "https://webservice-cw1.onrender.com/cities/41515"
```

`PUT /cities/{station_id}`

- **Purpose**: update city by station ID
- **Path param**:
  - `station_id` (string)
- **Body**: same fields as city create (without `station_id`)
- **Responses**:
  - `200`: updated city
  - `404`: city not found
  - `422`: validation failed

`DELETE /cities/{station_id}`

- **Purpose**: delete city by station ID
- **Path param**:
  - `station_id` (string)
- **Responses**:
  - `204`: deleted
  - `404`: city not found

4.3 Countries

`POST /countries`

- **Purpose**: create country record
- **Body**:
  - `iso2` (string, required, length 2)
  - `country` (string, required)
  - `native_name` (string, optional)
  - `iso3` (string, required, length 3)
  - `population` (number, optional)
  - `area` (number, optional)
  - `capital` (string, optional)
  - `capital_lat` (number, optional)
  - `capital_lng` (number, optional)
  - `region` (string, optional)
  - `continent` (string, optional)
- **Responses**:
  - `201`: created country
  - `409`: `iso2` already exists

- `422` : invalid body

Example:

```
curl -X POST "https://webservice-cw1.onrender.com/countries" \
  -H "Content-Type: application/json" \
  -d '{
    "iso2":"TS",
    "country":"Testonia",
    "native_name":"Testonia",
    "iso3":"TST",
    "population":1000000,
    "area":12345,
    "capital":"Test City",
    "capital_lat":10.1,
    "capital_lng":20.2,
    "region":"Test Region",
    "continent":"Asia"
  }'
```

`GET /countries`

- **Purpose**: list countries with pagination and optional continent filter
- **Query params**:
  - `skip` (int, default `0` , min `0`)
  - `limit` (int, default `20` , min `1` , max `300`)
  - `continent` (string, optional)
- **Response**: `200` + country array

Example:

```
curl "https://webservice-cw1.onrender.com/countries?
skip=0&limit=10&continent=Asia"
```

`GET /countries/{iso2}`

- **Purpose**: get country by ISO2 code
- **Path param**:
  - `iso2` (string)
- **Responses**:
  - `200` : country object
  - `404` : country not found

Example:

```
curl "https://webservice-cw1.onrender.com/countries/AF"
```

`PUT /countries/{iso2}`

- **Purpose**: update country by ISO2 code
- **Path param**:
  - `iso2` (string)
- **Body**: same fields as country create (without `iso2`)
- **Responses**:
  - `200`: updated country
  - `404`: country not found
  - `422`: validation failed

`DELETE /countries/{iso2}`

- **Purpose**: delete country by ISO2 code
- **Path param**:
  - `iso2` (string)
- **Responses**:
  - `204`: deleted
  - `404`: country not found

4.4 Analytics

`GET /analytics/cities-per-country`

- **Purpose**: count cities per country and sort descending
- **Query params**:
  - `limit` (int, default `10`, min `1`, max `100`)
- **Response**: `200` + array
  - `iso2`
  - `country`
  - `city_count`

Example:

```
curl "https://webservice-cw1.onrender.com/analytics/cities-per-country?limit=10"
```

`GET /analytics/countries-by-continent`

- **Purpose**: count countries by continent
- **Query params**: none
- **Response**: `200` + array
  - `continent`
  - `country_count`

Example:

```
curl "https://webservice-cw1.onrender.com/analytics/countries-by-continent"
```

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5. Assessment Submission Notes

- Keep this file as the detailed Markdown API documentation.
- Export this file to PDF as: `docs/API_Documentation.pdf`.
- Ensure `README.md` references the PDF file path.